

AY25 PROGRAM ASSESSMENT REPORT

v2026.1

Tracking Data

CIM code (e.g., ME-CM)	MS-AM
Program Name	Master of Science in Advanced Manufacturing
Assessment Year	2025
Program Assessment Coordinator	Heng Wang
Year of Current Assessment Plan	2024
No Assessment This Year per Plan	☐
LO(s) Assessed	LO1, LO2 (data collection and archiving only)
Presented at program authority meeting date: <i>Please attach minutes</i>	TBD

Methodology

LO#1

- ⌘ Learning goal assessed: Graduates should have a thorough understanding of traditional manufacturing technologies.
- ⌘ Sample size: All students completing MMAE 546, MMAE 547, MMAE 560, MMAE 557
- ⌘ Semesters in which artifacts were collected: Fall, Spring
- ⌘ Artifact: For each course, performance of the students in 4 questions chosen from a pool of questions specifically designed to assess this learning goal. The questions are delivered to the students in the last third of the semester. Each student will be scored out of 4.
- ⌘ Proficiency ratings
 - ⌘ scores of >75% points indicate **P** proficient.
 - ⌘ scores of >55% points indicate **M** meets expectations.
 - ⌘ scores below 55% points indicate **D** does not meet expectations.

LO#2

- ⌘ Learning goal assessed: Graduates should have a thorough understanding of evolving manufacturing technologies.
- ⌘ Sample size: All students completing a subset of MMAE 546, MMAE 588
- ⌘ Semesters in which artifacts were collected: Fall, Spring
- ⌘ Artifact: For each course, performance of the students in 4 questions chosen from a pool of questions especially designed to assess this learning goal. The questions are delivered to the students in the last third of the semester. Each student will be scored out of 4.

- ⊄ Proficiency ratings
 - ⊄ scores of >75% points indicate **P** proficient.
 - ⊄ scores of >55% points indicate **M** meets expectations.
 - ⊄ scores below 55% points indicate **D** does not meet expectations.

LO#3

- ⊄ Learning goal assessed: Graduates should have a thorough understanding of the relationships between raw materials, manufacturing processes, and product properties.
- ⊄ Sample size: All students completing a subset of MMAE 588
- ⊄ Semesters in which artifacts were collected: Fall, Spring
- ⊄ Artifact: For each course, performance of the students in 4 questions chosen from a pool of questions especially designed to assess this learning goal. The questions are delivered to the students in the last third of the semester. Each student will be scored out of 4.
- ⊄ Proficiency ratings
 - ⊄ scores of >75% points indicate **P** proficient.
 - ⊄ scores of >55% points indicate **M** meets expectations.
 - ⊄ scores below 55% points indicate **D** does not meet expectations.

LO#4

- ⊄ Learning goal assessed: For graduates doing thesis, they will have the ability to conduct research, publish scientific findings, and disseminate that knowledge.
- ⊄ Sample size: All students completing their MS thesis.
- ⊄ Semesters in which artifacts were collected: Fall, Spring
- ⊄ Artifact: A specially designed questionnaire filled out by the thesis committee after the thesis defense exams. The committee members come to a consensus and rate the students on a scale of 1-5 on each of the following:
 - ⊄ Research problem is clearly described
 - ⊄ Literature and context of the work are demonstrated
 - ⊄ Solution approach, methods and/or tools are described
 - ⊄ Results are analyzed, interpreted and communicated well
 - ⊄ Written and oral communication are effective
 - ⊄ Critical thinking consistent with an independent researcher
- ⊄ Proficiency ratings
 - ⊄ scores of 4 points and above for all items indicate **P** proficient.
 - ⊄ scores of 3 points and above for all items indicate **M** meets expectations.
 - ⊄ scores below 3 points on any item indicate **D** does not meet expectations.

Results

Artifact samples

Artifact for LO1 from MMAE 546. Four questions from a bank of five were selected and assigned as a quizz. Sample scores listed below. “C” for correct, “I” for incorrect.

1 Multiple Choice 1 point Which of the following is a primary advantage of powder metallurgy over traditional casting methods? ☒ Ability to produce complex shapes with high dimensional accuracy ☐ Lower energy consumption ☐ Better surface finish ☐ Lower material waste MS-AM LO1	3 Multiple Choice 1 point In traditional machining processes, which of the following factors most significantly affects tool life? ☒ Cutting speed ☐ Feed rate ☐ Depth of cut ☐ Workpiece material hardness MS-AM LO1
2 Multiple Choice 1 point In the context of traditional welding processes, what is the main purpose of flux in arc welding? ☒ To protect the weld pool from atmospheric contamination ☐ To increase the welding speed ☐ To reduce the temperature of the weld pool ☐ To increase the strength of the weld MS-AM LO1	4 Multiple Choice 1 point What is the primary advantage of using numerical control (NC) in traditional manufacturing processes? ☒ Improved consistency and accuracy of parts ☐ Reduced need for skilled operators ☐ Increased production speed ☐ Lower initial equipment costs MS-AM LO1

	Sample 1		
P	C, C, C, C (100%)		
M	C, C, C, I (75%)		
D	C, I, I, C (50%)		

Artifact for LO2 from MMAE 546. Four questions from a bank of five were selected and assigned as a quizz. Sample scores listed below. “C” for correct, “I” for incorrect.

1 Multiple Choice 1 point

In the context of Industry 4.0, what is the primary function of the Internet of Things (IoT) in manufacturing?

- ☒ To enable real-time monitoring and control of production processes
- ☐ To replace human workers with robots
- ☐ To eliminate the need for quality control
- ☐ To reduce energy consumption in all manufacturing processes

MS-AM LO2

2 Multiple Choice 1 point

What is a key advantage of using water jet cutting in modern manufacturing?

- ☒ It produces no heat-affected zone in the workpiece
- ☐ It can cut any material regardless of hardness
- ☐ It is faster than laser cutting for all materials
- ☐ It requires no post-processing of cut edges

MS-AM LO2

3 Multiple Choice 1 point

In the context of laser-based manufacturing, what is the primary benefit of using fiber lasers over CO2 lasers?

- ☒ Higher absorption in metals
- ☐ Lower initial equipment cost
- ☐ Larger spot size for wider cuts
- ☐ Better performance on non-metallic materials

MS-AM LO2

4 Multiple Choice 1 point

Which of the following is a key advantage of additive manufacturing (3D printing) over traditional manufacturing methods?

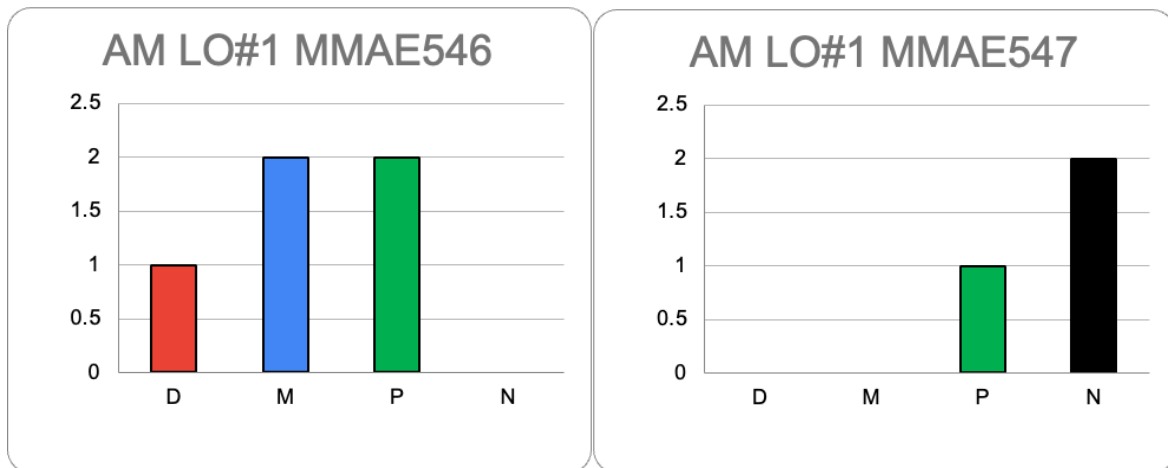
- ☒ Ability to create complex geometries without tooling
- ☐ Higher production speed for large batch sizes
- ☐ Lower energy consumption for all part sizes
- ☐ Better surface finish without post-processing

MS-AM LO2

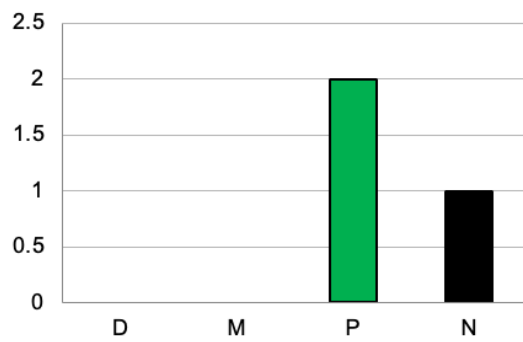
	Sample 1		
P	C, C, C, C (100%)		
M			
D			

Data table

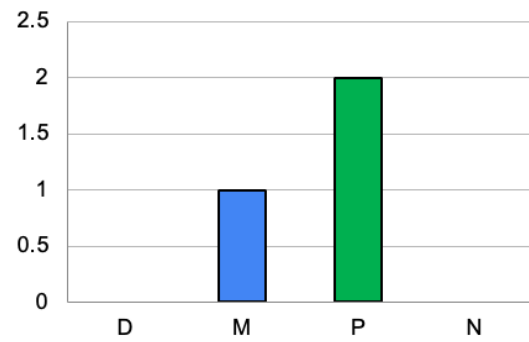
Charts below show the distribution of results from different courses. Y axis is the number of students.



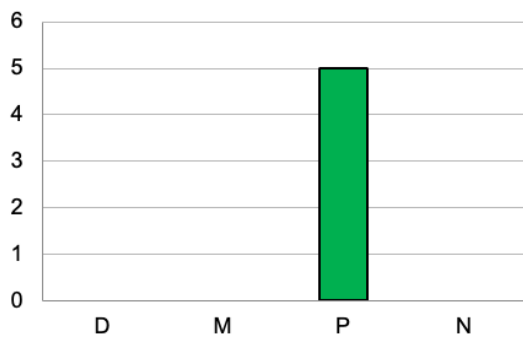
AM LO#1 MMAE560



AM LO#1 MMAE557



AM LO#2 MMAE546



Description of findings

Due to insufficient sample size of students. The department has decided to only collect and archive artifacts from students. Analysis will be performed when sufficient data is collected. Based on what has been collected this year no concern was recognized.

Continuous Improvement, CI, plans

The purpose of assessment is to improve the quality of our educational programs. The most important part of the assessment cycle is how we use the data to improve our programs. Examine the data provided above describing student achievement, consider the root cause of any deficits, and if areas for possible improvement emerge, describe plans to achieve this improvement.

Prior CI plan impact

Also, please refer to the previous assessment report for the LOs assessed. Were any CI plans recommended? Were they implemented? Did the plans work (were improvements seen in this cycle)?

☒ no previous CI plans available

This cycle's CI plan

Describe any CI plans developed from this cycle's data, with

- Specific curricular or instructional changes in response to this assessment data,
- A timeline for implementation,
- As well a description of who will participate in and manage these plans

☒ After consideration and presentation to the program authority, no CI plans are recommended.